Translated by DeepSeek

Summary and Outlook

This paper begins with a small incident in daily life—a series of persistent questions raised by my son while I was tutoring him in physics during his second year of junior high school—which triggered a prolonged meditation spanning multiple levels of physics. Faced with many fundamental and unresolved puzzles in modern physics, we attempt to shift our perspective and re-understand the world through more fundamental and simpler first principles.

Our core hypothesis starts with a simple yet bold conjecture: the proton is not extraordinarily stable as we observe; on the contrary, like other ordinary particles, its lifetime is extremely short—only because we ourselves are made of protons are we inherently unable to perceive its decay. This is precisely the physical embodiment of the classical Chinese poem, "Not knowing the true face of Mount Lu, simply because one is standing in its midst."

From this starting point, we gradually construct a logically self-consistent theoretical framework, whose core can be summarized in the following seven steps:

1. **Annihilation-Rebirth Mechanism**: The proton is not a continuously existing entity; rather, it cyclically undergoes annihilation and identical regeneration on an extremely short timescale. Before and after each birth-death cycle, the particle is identical, and the energy budget is strictly balanced, so observers are inherently trapped within this rhythm.
2. **Observer Limitation**: Our perceptual channels are almost entirely monopolized by electromagnetism and gravitation, both of which tacitly assume the "persistent existence of the observed object" as a working premise. It is not that protons do not perish, but that we lack the capacity to witness their perishing.
3. **World Phase and Electromagnetic Phase**: Treating the proton's complete birth-death cycle as one period (2π), the World Phase corresponds to the rebirth cycle, while the Electromagnetic Phase is an observation window within the World Phase. The Electromagnetic Phase delimits the boundaries of our cognition, and the appearance of the World Phase is presented by the Electromagnetic Phase.
4. **Pure AC Energy Model**: Quantum fluctuation is the fundamental form of the universe's energy operation; the sum of energy across all World Phases in the universe is identically zero. The borrowing and returning of energy in positive and negative half-cycles link end to end, forming a self-sustaining cycle. The

initial energy of the Big Bang is nothing more than a replaceable DC bias.

5. **Phase Interpretation of Identical Particles**: All particles are essentially the same entity (wave packet), and their differing apparent properties arise solely from their different phase positions within the World Phase—the proton's stability comes from its central phase position, the electron's dual nature from its position at the edge of the World Phase, and the photon is the trace of energy settlement when a particle crosses World Phases during rebirth.
6. **Void Substrate and Void Breaking**: The platform that gives birth to identical particles is ultimately founded upon nothingness—the "nothing" of philosophy, the entity corresponding to the empty set (∅) in mathematics. The non-empty power set of the empty set inherently contains the possibility of "something." The void breaking is the natural result of the void imposing no prohibitions. The non-locality of the breaking naturally possesses uniformity and isotropy, replacing the function of inflation theory.
7. **Spacetime Lattice and Three-Valued Resonance**: The void breaking is described by the impulse sequence {+1, 0, -1, 0}; matter and space are two sides of a single coin. Through the three-stage evolution of chaos → resonance → stability, stable physical parameters emerge in the stable period, and time and space are thereby born. The motion of matter is a relay of annihilation and rebirth on a discrete lattice—"Matter tells space how to rebirth; space tells matter how to move."

Through systematic examination, this "primordial broth"—the void breaking and the discrete dynamics of the spacetime lattice—not only satisfies but, at key points, surpasses the initial postulates of quantum mechanics and quantum field theory: postulates such as identical particles, fields as fundamental entities, microscopic causality, and vacuum fluctuations acquire an ontological foundation, no longer being mere empirical inputs; the measurement collapse and the probabilistic origin of the Born rule are explained by deeper mechanisms, no longer independent postulates; the infinities in renormalization are eliminated by the natural ultraviolet cutoff of the discrete lattice, and the origin of gauge symmetry can be derived from the local consistency of lattice update rules.

Based on the above framework, we propose a unified ontological explanation for several core puzzles in quantum mechanics: Hilbert space is a faithful mapping of the phase configurations within the World Phase; wave function collapse is the synchronized resonance of lattices within the Electromagnetic Phase window; the uncertainty principle is rooted in the discreteness of the spacetime lattice; quantum entanglement is a natural remnant of the non-locality of the void breaking; wave-particle duality represents two complementary perspectives on the same discrete propagation process; quantum tunneling is the probabilistic jumping of annihilation and rebirth between adjacent lattices; and vacuum fluctuations are the residual local breakings on the lattice not yet fully synchronized in resonance.

Of course, substantial work remains to be done in progressing from qualitative conjecture to quantitative calculation. Redefining field operators on the discrete lattice, deriving propagators, establishing Feynman rules, and verifying that the effective field theory form of standard quantum field theory is recovered in the long-wave limit will be the crucial test for whether this framework can truly take root.

Nevertheless, we have already taken the first step: starting from a simple question, relying on physical intuition and repeated deep contemplation, we have constructed a logically self-consistent picture of the universe rooted in first principles. The world should not be this complicated—and perhaps, it really is not as complex as we have imagined.

中国的深度求索人工智能对论文的总结

总结与展望

本文从一件日常生活的小事出发——为读初二的儿子辅导物理时的一连串追问——触发了一场跨越多个物理学层级的沉思。面对现代物理学中诸多悬而未决的根本性疑难，我们尝试换一个角度，用更根本、更简单的第一性原理来重新理解世界。

我们的核心假设始于一个朴素而大胆的猜想：质子并非如我们观测到的那样异常稳定，恰恰相反，它和其他普通粒子一样寿命极为短暂——只是因为我们自身由质子构成，才天然地无法感知它的衰亡。这正是“不识庐山真面目，只缘身在此山中”的物理写照。

由此出发，我们逐步构建了一个逻辑自洽的理论框架。其核心可概括为以下七个环节：

1. **湮灭重生机制**：质子并非持续存在的实体，而是在极短时间尺度上循环经历湮灭与全同再生，每一次生灭前后粒子全同、能量收支严格平衡，观测者天然被困于这一节律之中。
2. **观测者局限**：我们的感知通道被电磁和引力所垄断，两者都以“探测对象的持续存在”为不言自明的工作前提。不是质子不灭，而是我们没有能力看见它的灭。
3. **世界相与电磁相**：将质子的生死循环视为一个周期（2π），世界相对应重生周期，电磁相是世界相内部的一个观测窗口。电磁相框定了我们的认知边界，世界相的表象由电磁相所呈现。
4. **纯交流能量模型**：量子涨落是宇宙能量运转的基本形式，宇宙中所有世界相的能量总和恒为零，正负半周的能量借还与归还首尾相衔，构成自持的循环。大爆炸的初始能量不过是可被替换的直流偏置。
5. **全同粒子的相位解释**：所有粒子本质上是同一实体（波包），仅因在世界相中的相位位置不同而呈现出不同的表观性质——质子稳定因其居于中心相位，电子的双重性因其处于世界相边缘，光子则是粒子跨世界相重生时能量结算的痕迹。
6. **虚无基底与虚空破缺**：全同粒子诞生的平台最终建立在虚无之上——哲学上的“无”，数学上空集（∅）所对应的本体。空集的非空幂集内蕴了“有”的可能性，虚空破缺是虚无不设禁的自然结果。破缺的非局域性天然具备均匀性与各向同性，替代了暴涨理论的功能。
7. **时空晶格与三值谐振**：虚空破缺由冲激序列{+1, 0, -1, 0}描述，物质与空间乃一体两面。经混沌→共振→稳定三阶段演化，稳定的物理参数在稳定期涌现，时间与空间得以诞生。物质的运动是离散晶格上的湮灭重生接力——“物质告诉空

间如何重生，空间告诉物质如何运动”。

经系统审查，这一“底汤”——虚空破缺与时空晶格离散动力学——不仅满足，而且在关键点上超越了量子力学和量子场论的初始公设：全同粒子、场为实体、微观因果性、真空涨落等公设获得了本体论基础，不再是单纯的经验输入；测量坍缩、玻恩规则的概率来源被更底层的机制所解释，不再是独立公设；重整化中的无穷大被离散晶格的自然紫外截断所消解，规范对称性的来源可在晶格局域一致性中导出。

基于上述框架，我们对量子力学中若干核心疑难提出了统一的本体论解释：希尔伯特空间是世界相内相位配置的忠实映射，波函数坍缩是电磁相窗口内的晶格共振同步，不确定性原理根源于时空晶格的离散性，量子纠缠是虚空破缺非局域性的自然遗存，波粒二象性是离散传播过程的两个互补视角，量子隧穿是湮灭重生在晶格间的概率性跳变，真空涨落则是晶格上未完全共振同步的局域破缺残余。

当然，从定性猜想到定量计算之间，仍有大量工作待完成。在离散晶格上重新定义场算符、导出传播子、建立费曼规则，并在长波极限下验证标准量子场论的有效场论形式得以恢复，将是这一框架能否真正立足的关键考验。

但无论如何，我们已经迈出了第一步：从一个朴素追问出发，凭借物理直觉与反复深思，构建了一个逻辑自洽、根植于第一性原理的宇宙图景。世界本不该那么复杂——或许，它真的没有我们想象中那么复杂。